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Highlights

- We develop a model that coordinates between dynamic pricing and inventory decisions.
- Customer demand is assumed to be additive in price and time since replenishment.
- We examine changes in price when approaching the optimal replenishment time.
- Nonlinear effects of price and time on demand provide insights into optimal pricing.
- We evaluate benefit from the implementation of a dynamic pricing policy.

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Optimal dynamic pricing and ordering of a perishable product under additive effects of price and time on demand

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Abstract

When perishable products, such as dairy products, fruits and vegetables, drugs, or batteries are priced uniformly, without taking into consideration the amount of time remaining until the expiration date, consumers may gravitate towards fresher products, leaving some inventory unsold. A dynamic pricing policy, in which products are priced differently as they approach expiration, may encourage customers to buy less-fresh products, potentially increasing revenue and eliminating waste. Following scarce literature on dynamic pricing of storable perishable items, this paper develops a model to determine a product's optimal replenishment schedule and dynamic price over time, with the aim of maximizing the retailer's profit. Customer demand is assumed to be a pseudo-additive function of price and time since replenishment. Some properties of the optimal pricing and replenishment policy are derived by means of necessary and sufficient conditions of optimality. A number of examples show the dynamics of the optimal policy under different assumptions regarding demand. In particular, we evaluate the extent to which the retailer can benefit from the implementation of a dynamic pricing policy as opposed to a static one, and we show that the optimal policy is highly dependent on the form of demand incorporated into the model.

Key words

Inventory; Dynamic pricing; Differential pricing policy; Additive demand; Perishable items

1. Introduction

Dynamic pricing is an efficient revenue management policy that is applied to numerous products and services, particularly in situations in which managers must sell a given stock of items by a deadline. This is the case for perishable products, i.e., products that become worthless if they are not sold by a specific time (the expiration date; Fleischmann *et al*., 2004). Some dynamically-priced perishable products - e.g., products in the travel and leisure industry, such as seats on airline flights, cabins on vacation cruises, and rooms in hotels (Gallego and van Ryzin, 1994) - are not held in inventory, meaning that they are not associated with holding costs or fixed ordering costs, and their demand does not necessarily

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decrease with time. Inventoried perishable products, in contrast, are discarded after their expiration dates and are associated with constant holding and ordering costs. Examples of these products include dairy products, fruits and vegetables, beverages, drugs, and batteries. Demand for inventoried perishables, given a fixed price, decreases over time, as expiration approaches.

This paper focuses on inventoried perishable products and develops an inventory replenishment model that aims to coordinate between dynamic pricing and inventory decisions. We consider an inventory system that consists of a single type of storable perishable product, which is periodically replenished under a demand that depends on price and on products' "freshness", i.e., time elapsed since replenishment. Our objective is to develop an optimization model to dynamically determine the price, to schedule replenishments, and to set their quantities, such that the retailer's profit is maximized.

1.1 Dynamic pricing of inventoried perishable items

Dynamic pricing of inventoried perishable items is a common practice, and can be carried out both in shops and online. In many cases, managers update prices automatically, an approach that is less costly and easier to implement compared with manual price adjustment (Chan *et al*., 2004; Odlyzko, 2004). Such automation can be achieved using technologies such as RFIDs.

For a review of the literature and current practices on dynamic pricing in the presence of inventory considerations, the reader is referred to Elmaghraby and Keskinocak (2003) and to Chan *et al*. (2004). A review of the relevant literature on the problem of setting prices for perishable inventories can be found in the first section of Zhao and Zheng (2000). Studies that refer to specific scenarios of dynamic pricing include the work of Gupta *et al*. (2006), which presents a discrete-time model for setting clearance prices for retail inventories of fashion goods. A study by Wee and Yu (1997) considers the effects of a temporary discount sale when the items deteriorate exponentially with time. Additional papers that discuss dynamic pricing of storable products include Levin *et al*. (2010), Chung *et al*. (2007), Roy and Chaudhuri (2007), Yang (2004), Kang and Kim (1983), and Smith (1975).

1.2 Modeling the combined effects of price and time on demand

In the presence of competition, inventory-based businesses are forced to coordinate their procurement decisions with their marketing decisions, in order to avoid carrying excessive stock when sales are low, or shortages when they are high (Polatoglu and Sahin, 2000). As

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demand for perishable products is affected both by price and by the remaining time until products' expiration, it is essential to model these effects simultaneously in order to establish an optimal pricing and inventory replenishment policy. Yet few research efforts have explicitly considered the dependency of demand on both price and time. Urban and Baker (1997) study a model in which demand is a deterministic, multivariate function of price, time, and inventory level. A review of deterministic models of inventory-level-dependent demand appears in Urban (2005). Wu *et al*. (2006) extended Urban and Baker's (1997) model to include non-instantaneous deterioration of items, and Devangan *et al*. (2013) extended the model to stochastic demand. Models that explicitly address the dependency of demand on price and time appear in Valliathal and Uthayakumar (2011), Maihami and Kamalabadi (2012), and Avinadav *et al*. (2013). These works assume that demand is dependent on a multiplicative combination of price and time and that price is static over the time horizon.

1.3 Results of related works

When modeling a demand rate that is affected by both price and remaining time until expiration, one may assume either a multiplicative relation or an additive relation between the effects. Herbon (2014) used a multiplicative representation of demand to model an inventory system in which a perishable product is periodically replenished, and in which consumers are heterogeneous in their sensitivity to product freshness—although the retailer is unaware of this heterogeneity. That paper shows that the price at late stages of the product's shelf-life should be higher than the price at early stages of the shelf-life. Similarly to Avinadav *et al*. (2013), Herbon (2014) assumes that the demand rate decreases linearly in the selling price and polynomially over the time after replenishment, until it vanishes either at the reservation price or at the time of expiration. A two-level piecewise constant function was proposed to represent the pricing policy, and an optimal solution was obtained.

You (2005) models demand for perishable products on the basis of an additive combination of price and time. Specifically, the author studies a discrete-in-time dynamic pricing policy, where demand is an additive combination of a linear price factor and an exponential time factor. Liu *et al*. (2008) presents a closed-form solution for the particular case of continuous-time demand linear in price and exponentially decayed in time. Avinadav *et al*. (2014) address an additive demand model with a fixed price and show that the separability of price and time leads to inconsistencies with the practice of marketing perishable products. In light of this observation, they suggest a pseudo-additive model and

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illustrate its applicability to the case of demand that decreases linearly with price. In this paper, we generalize the definition of pseudo-additive demand and show its applicability to a larger class of demand models, as well as to a dynamic pricing environment.

1.4 Contribution of the current research

As noted above, this paper develops an inventory replenishment model with dynamic pricing, and with demand that depends on price and time elapsed from replenishment. It presents a general framework for analyzing the problem, special cases of which have been discussed in Section 1.3; these cases demonstrate the problem's theoretical and practical relevance. In particular, we seek to address the following gaps in the literature:

(1) Generalizing the pseudo-additive model suggested in Avinadav *et al*. (2014) to be applicable to dynamic pricing and to a larger class of demand models, such as non-linear dependency of the effect of price on demand.

(2) Studying the properties of the optimal policy, specifically, properties that are not contingent on the separability of the price and time effects, and properties that are.

(3) Obtaining the optimal dynamic pricing policy for the model we develop.

(4) Showing how the retailer's optimal policy under a specific form of the (pseudo-additive-demand-based) model developed herein differs from his optimal policies under different forms of demand (either multiplicative or additive).

(5) Comparing the implementation of continuous dynamic pricing in deterministic and stochastic environments.

(6) Generalizing the model to adopt a differential pricing policy.

In the next section we formulate the model and the assumptions, and present properties of the optimal policy. Section 3 develops the solution method and numerical algorithm. Sections 4 and 5 analyze the optimal policy under linear and non-linear effects of price on demand, respectively. Section 6 discusses implementation issues. In particular, it numerically exemplifies and emphasizes the differences between the additive dynamic model as compared with additive and multiplicative demand models with a fixed price, for the special case in which the demand decreases linearly in the selling price and polynomially over the time after replenishment. Section 7 expends the model to the case of a differentiable pricing policy. Section 8 concludes the paper.

2. Problem formulation

We model a retailer who sells a single perishable item, without shortage, where batch replenishment is performed periodically at the retailer's discretion. In addition to the common decision variables of when and how much to order, we also consider the selling price as a decision variable; this price is dynamically determined along the replenishment period.

Notation:

Parameters: K – the fixed ordering cost per replenishment.

c – the purchasing cost per unit.

h – the unit inventory holding cost per time unit.

$T_{\max}$ – the duration until expiration after replenishment.

Decision variables: Q – order quantity.

$p(t)$ – the selling price of the item at time t.

T – the replenishment period.

Variables: t – the time elapsed from the last replenishment.

$\lambda(p(t),t)$ – the demand rate at t for a given $p(t)$.

$I(Q,p(t),t)$ – the inventory level at t for given $p(t)$ and Q.

The profit per replenishment period, $\Pi(Q,p(t),T)$, includes three components: the net profit gained from sold units; the total inventory holding costs between replenishments; and the fixed ordering cost per replenishment. Thus,

$$\Pi(Q,p(t),T)=\int_0^T (p(t)-c)\lambda(p(t),t)dt-h\int_0^T I(Q,p(t),t)dt-K. \tag{1}$$

The objective function is the average profit per unit of time,

$$\pi(Q,p(t),T)\equiv\Pi(Q,p(t),T)/T, \tag{2}$$

so that the optimization problem addressed in this paper is

$$\max_{Q,p(t),T}\{\pi(Q,p(t),T)\}. \tag{3}$$

The following assumption specifies the demand rate function, $\lambda(p(t),t)$.

Assumption 1. The demand rate $\lambda(p,t)$ is non-negative, $\lambda(p,t)\geq 0$, and explicitly and separately depends on both the price factor and time factor, according to an additive combination of functions $\lambda_1(p)$ and $\lambda_2(t)$, respectively,

$$\lambda(p,t)=\lambda_1(p)+\lambda_2(t). \tag{4}$$

The function $\lambda_1(p)$ is strictly decreasing and differentiable.

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We note that since the selling price, $p(t)$, may change in time, which is the case studied in this paper, the dependence of demand on time is two-fold: explicit through the time factor $\lambda_2(t)$, and implicit through the dynamic pricing $\lambda_1(p(t))$.

A general form of (4) requires further specification, since, in general, it is inconsistent with the requirement that when $p(t)$ tends to infinity, the demand rate must vanish, i.e.,

$$\lim_{p(t)\to\infty} \lambda(p(t),t) = 0 \text{ for all } t. \tag{5}$$

The inconsistency is shown by combining (4) and (5), and follows from the fact that $\lambda_2(t) = -\lim_{p(t)\to\infty} \lambda_1(p(t))$ is constant. This contradicts Assumption 1, which requires the existence of a time effect on demand. In order to come up with a consistent definition of $\lambda(p(t),t)$, it is necessary to incorporate an additional assumption, which characterizes the marketing of perishable products.

Assumption 2. There exists a reservation price (Steedman, 1987), $p_{\max}(t)$, defined as the lowest price at which the product is no longer demanded at *t*. The reservation price satisfies the following conditions,

(i) $p_{\max}(t) = 0$ for $t \geq T_{\max}$;

(ii) $\lambda_1(p_{\max}(t)) + \lambda_2(t) = 0, \quad 0 \leq t \leq T_{\max}$ (6)

(iii) $p_{\max}(0) > c$ and $p_{\max}(t)$ continuously decreases over $0 \leq t \leq T_{\max}$.

From Assumption 2 we obtain the following.

Lemma 1. The function $\lambda_2(t)$ is strictly decreasing.

Proof. By differentiating (6) with respect to *t*, we obtain

$$\frac{d\lambda_1(p_{\max})}{dp_{\max}} \frac{dp_{\max}(t)}{dt} + \frac{d\lambda_2(t)}{dt} = 0.$$

The lemma follows, since $\frac{dp_{\max}(t)}{dt} < 0$ over $0 \leq t \leq T_{\max}$ and $\frac{d\lambda_1(p)}{dp} < 0$. □

From Assumption 2 it also immediately follows that no demand appears beyond $T_{\max}$ regardless of $p(t)$, i.e., $\lambda(p(t),t) = 0$ for $t \geq T_{\max}$, and that the value of $p_{\max}(t)$ is calculated by,

$$p_{\max}(t) = \lambda_1^{-1}(-\lambda_2(t)). \tag{7}$$

Note that although the conditions of Assumption 2 hold true for the practice of perishable products, they might not be satisfied for some non-inventoried products, such as

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cruise or concert seats, or for very unique inventoried products such as wine or art objects. In particular, the reservation price $p_{\max}(t)$ in such cases might grow in time.

Assumption 2 resolves the inconsistency between (4) and (5). The additive demand rate, defined as

$$\lambda(p(t),t) = \begin{cases} \lambda_1(p(t)) + \lambda_2(t) & \text{for } t \le T_{\max} \text{ and } p(t) \le p_{\max}(t) \\ 0 & \text{otherwise} \end{cases}, \tag{8}$$

completes the statement of problem (3).

Definition (8) extends the concept of a *pseudo-additive* demand rate, discussed in Avinadav *et al*. (2014), to the case of dynamic pricing. In contrast to the additive demand model in You (2005), the demand in (8) never becomes negative. In contrast to the multiplicative demand model (see, e.g., Valliathal and Uthayakumar, 2011 and Avinadav *et al*. 2013), in which each of the two effects of demand must be non-negative, this is not necessarily true in (8) as long as the total $\lambda(p(t),t)$ remains non-negative. Section 4 further discusses and exemplifies the properties of (8).

3. Analysis of the model

3.1 General properties that do not require separability

This section provides properties of an optimal solution of (3) that have a more general scope and are not contingent on the separability of demand into components of price and time.

First, it is reasonable to bound the replenishment period, *T*, by the expiration period, i.e.,

$$T \le T_{\max}. \tag{9}$$

This is because no demand occurs beyond $T_{\max}$, and no profit is obtained beyond this point.

Second, we note that for any *p*(*t*), no inventory remains at the replenishment point, *T*, $I(Q, p(t), T) = 0$. This conclusion is in line with that of some other inventory models (see, e.g., Lui *et al*. (2008)) and is proven in the following lemma. The proof of the lemma is based on the assumption of an equal pricing policy, i.e., a policy in which all items on the shelf are priced identically. The case of a differential pricing policy, in which different prices are assigned to freshly-delivered items and to items remaining from the previous replenishment period is considered in Section 7.

Lemma 2. No inventory remains at the end of the replenishment cycle, $I(Q, p(t), T) = 0$.

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Proof. Suppose that the pricing policy $p(t)$ and quantity Q result in $I(Q, p(t), T) = q > 0$. Consider this same policy, $p(t)$, applied over the same period $t \in [0, T]$, with the initial inventory decreased by q, i.e., when Q is substituted with $Q - q$. This yields a new inventory time path $I(Q - q, p(t), t)$ and a new objective value $\pi(Q - q, p(t), T)$. Since the demand function $\lambda(p, t)$ does not depend on the level of inventory, but rather on price and time, the new demand is identical to that observed when applying the original policy. Therefore,

$$I(Q - q, p(t), t) = I(Q, p(t), t) - q \quad \text{for } 0 \le t \le T .$$

From (1) and (2) the new objective is

$$\pi(Q - q, p(t), T) = \frac{1}{T}\left(\int_0^T (p(t) - c)\lambda(p(t), t)dt - h\int_0^T I(Q - q, p(t), t)dt - K \right) =$$

$$= \frac{1}{T}\left(\int_0^T (p(t) - c)\lambda(p(t), t)dt - h\int_0^T \big(I(Q, p(t), t) - q\big)dt - K \right) =$$

$$= \pi(Q, p(t), T) + hq .$$

The lemma now immediately follows, since the new solution is better than the previous one for any $q > 0$. □

Third, the order quantity Q is not an independent decision variable. It depends on the other decision variables $p(t)$ and T. This follows from Lemma 2, since Q, as proven, is the total demand aggregated over the replenishment period.

Fourth, the optimization problem can be solved hierarchically, such that at a lower level of the hierarchy, optimal price and inventory profiles are obtained for a given T. Then, at a higher hierarchical level, T_{opt} is obtained by,

$$\max_{p(t),T} \{\pi(p(t), T)\} = \max_{T \le T_{\max}} \left\{ \max_{p(t) \le p_{\max}(t)} \{\pi(p(t), T)\} \right\}. \tag{10}$$

Sections 3.2 and 3.3 below solve the optimization problems at the two hierarchical levels. The sequential hierarchical optimization, first with respect to the decision variable $p(t)$, and then with respect to the decision variable T, covers all possible combinations of the two decision variables. Therefore, the objective $\pi(p(t), T)$ is correctly maximized.

3.2 The optimal dynamic price given the replenishment period

Since $\max_{p(t)}\{\pi(p(t),T)\} \equiv \frac{1}{T}\max_{p(t)}\Pi(p(t))$, the problem at the lower hierarchical level of the decomposition discussed above is stated as follows: Given a replenishment period, *T*, identify a dynamic price function $p(t)$ that maximizes the total profit for the period,

$$\max_{p(t)}\Pi(p(t)) = \max_{p(t)}\left\{\int_0^T \big((p(t)-c)\lambda(p(t),t) - hI(t)\big)dt - K\right\} \tag{11}$$

s.t.

$$\frac{d}{dt}I(t) = -\lambda(p(t),t), \quad I(T)=0,\ 0 \le t \le T, \tag{12}$$

$$0 \le p(t) \le p_{\max}(t).$$

The optimal control problem (11)-(12) can be solved by means of the maximum principle (Hartl *et al*., 1995, Sethi and Thompson, 2000), which states that if $p_{opt}(t)$ solves the problem, then there exists a costate function $\psi(t)$ that satisfies the costate equation,

$$\frac{d}{dt}\psi(t) = -\frac{\partial H}{\partial I} = h, \quad \psi(0)=0, \tag{13}$$

where the Hamiltonian,

$$H = (p(t)-c)\lambda(p(t),t) - hI(t) - \psi(t)\lambda(p(t),t) \tag{14}$$

is maximized by $p_{opt}(t)$ within the feasible domain, $0 \le p(t) \le p_{\max}(t)$. The boundary costate condition $\psi(0)=0$ in (13) follows from the transversality conditions that link between the state and costate boundaries. In our case, the transversality conditions require that since the initial state, $I(0)$, is a decision variable, the initial costate is set at zero; since the terminal state is given, $I(T)=0$, the terminal costate, $\psi(T)$ is free.

The maximization of *H* is based on the solution of (13), $\psi(t) = ht$, and the fact that the factor $hI(t)$ appearing in the Hamiltonian, does not depend explicitly on *p*(*t*). As a result, the dynamic price function that solves the above optimal control problem is

$$p_{opt}(t) = \begin{cases} p_{\max}(t), & \text{if } \varphi(t) \ge p_{\max}(t), \\ \varphi(t), & \text{if } 0 \le \varphi(t) \le p_{\max}(t), \\ 0, & \text{if } \varphi(t) \le 0 \end{cases} \tag{15}$$

where $\varphi(t)$ is the root of the equation,

$$\lambda_1'(\varphi(t))(\varphi(t)-c-ht) + \lambda(\varphi(t),t) = 0, \qquad t \in [0,T]. \tag{16}$$

The second-order optimality conditions require that

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$$\frac{\partial^2 H}{\partial p^2} = 2\lambda_1'(p_{opt}(t)) + (p_{opt}(t) - c - ht)\lambda_1''(p_{opt}(t)) \le 0. \qquad (17)$$

Note that condition (17) further narrows the set of functions $\lambda_1(p)$ that are valid for the additive model, regardless of the function $\lambda_2(t)$.

The function $p_{opt}(t)$ is defined over the replenishment period, $t \in [0,T]$. However, the shape of $p_{opt}(t)$ does not depend on the length of the period. This follows since neither (15) nor (16) involves *T* explicitly. Given *T*, the dependence of the optimal price on time, *t*, can be equivalently reformulated into the dependence on the remaining time until replenishment, *T-t*. This property is quite intuitive, as it sets the price in accordance with the remaining time until a new period begins.

The next lemma puts $p_{opt}(t)$ in comparison to the two known functions of time, $c + ht$ and $p_{\max}(t)$.

Lemma 3. There exists a unique time T_1, such that

(i) $c + ht < p_{opt}(t) < p_{\max}(t)$ for $0 \le t < T_1$;

(ii) $c + ht = p_{opt}(t) = p_{\max}(t)$ for $t = T_1$;

(iii) $c + ht > p_{opt}(t) = p_{\max}(t)$ for $t > T_1$.

A specific case of the functions described in Lemma 3 is schematically plotted in Figure 1.

Proof. Since $p_{\max}(t)$ decreases in time, $p_{\max}(0) > c$, and $p_{\max}(t) = 0$ for $t > T_{\max}$ (Assumption 2), there exists a unique time denoted by T_1, such that $p_{\max}(t) > c + ht$ for $t < T_1$, and $p_{\max}(t) < c + ht$ for $t > T_1$.

If $\varphi(t) > p_{\max}(t)$, then according to the definition of the consistent demand rate (8), $\lambda(\varphi(t),t) = 0$, and after substituting that in (15) and (16), we obtain line (iii) of the lemma.

If $\varphi(t) < p_{\max}(t)$, then $\lambda(\varphi(t),t) > 0$. From (16) and Assumption 2 it now follows that $\varphi(t) > c + ht$, which proves line (i) of the lemma.

Line (ii) of the lemma just links the two time intervals – before and after T_1. □

If $T > T_1$, then the system does not operate between T_1 and T, since no demand and no inventory exist in this time interval. If $T \le T_1$, then inventory runs out at the replenishment point exactly, and the system operates for the entire duration of the replenishment period,

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with positive demand and inventory. This differs from the typical behavior of $p_{opt}(t)$ observed within multiplicative models (Herbon, 2014), where $p_{opt}(t)$ depends explicitly on *T*.

The dynamics of $p_{opt}(t)$ immediately follow from Lemma 3 and equation (16). The optimal dynamic price function has two modes corresponding to the first two lines of (15), respectively. It starts with mode 2 (second line of (15)) and switches to mode 1 (first line of (15)) at time T_1. When in mode 2, $p_{opt}(t)$ can either steadily grow, or steadily decrease, or have more complex behavior. After the switch to mode 1, $p_{opt}(t)$ equals $p_{\max}(t)$, and decays as a function of time (or equals zero). The observation that $p_{opt}(t)$ can grow over a period of time when in mode 2 is in line with the conclusions of You (2005) and of Herbon (2014), who solved different models of price-dependent demand. Likewise, this observation is in line with the work of Transchel and Minner (2009), who investigated an EOQ setting with price-dependent demand and showed that the optimal price is strictly increasing over time.

The optimal price function, $p_{opt}(t)$, determines the marginal profit, $r(t)$, gained from an item sold at time *t*,

$$r(t) = p_{opt}(t) - c - ht .$$

From (16) it follows that $r(t)$ can be expressed in terms of $p_{\max}(t)$ and $p_{opt}(t)$ as follows,

$$r(t) = -\frac{\lambda(p_{opt}(t),t)}{\lambda_1'(p_{opt}(t))} = \frac{\lambda_1(p_{\max}(t)) - \lambda_1(p_{opt}(t))}{\lambda_1'(p_{opt}(t))} \tag{18}$$

Equation (18) shows that $r(t)$ does not explicitly depend on the time effect on demand.

3.3 The optimal replenishment period

The problem at the higher hierarchical level of the decomposition discussed above is stated as follows: given the optimal dynamic price function, $p_{opt}(t)$, maximize the total profit under the replenishment period, *T*, as a single decision variable. We denote $\pi(T) = \pi(p_{opt}(t),T)$ and $\lambda(t) = \lambda(p_{opt}(t),t)$, and write

$$\max_T \pi(T), \qquad s.t. \quad T \le T_{\max} .$$

Section 3.2 has shown that there exists a time point T_1, such that beyond that point the optimal price equals $p_{\max}(t)$, and no demand can occur. Therefore, the optimal replenishment time must be no later than T_1,

$$T_{opt} \le T_1. \tag{19}$$

The next theorem determines three values that the optimal replenishment time, T_{opt}, can take.

Theorem 1.

If $\pi(T_1) > 0$, then $T_{opt} = \min\{T_2, T_{\max}\}$, where T_2 is a unique solution of the equation

$$\pi(T) = \lambda(T)(p_{opt}(T) - c - hT), \quad 0 \le T \le T_1. \tag{20}$$

Otherwise, no solution of (20) exists, and $T_{opt} = \min\{T_1, T_{\max}\}$.

Proof. We substitute $p_{opt}(t)$ in (2) and differentiate it with respect to T (see Appendix A for details)

$$\frac{d\pi(T)}{dT} = -\frac{1}{T}\Big(\pi(T) - \lambda(T)(p_{opt}(T) - c - hT)\Big). \tag{21}$$

Consider the derivative $\frac{d\pi(T)}{dT}$ on the interval $T \in [0, T_1]$. The second term in the right-hand side of (21), $\lambda(T)(p_{opt}(T) - c - hT)$, is positive and declines to zero at the end of the interval, $T = T_1$ (see Lemma 3). The first term, $\pi(T)$, is large and negative at the beginning of the interval, since $\lim_{T \to 0} \pi(T) = -\infty$. Therefore, from (21) we conclude that if $\pi(T)$ remains negative over the entire interval, then $\pi(T)$ constantly increases and reaches a maximum at $T = T_1$. This proves the second statement of the theorem. If $\pi(T)$ hits the zero level and becomes positive, then there must be an extreme point, T_2, so that $\left.\frac{d\pi(T)}{dT}\right|_{T=T_2} = 0$. This follows, since otherwise, $\pi(T_1)$ and $d\pi(T)/dT|_{T=T_1}$ do not fit each other. To show the uniqueness of T_2, we consider the second derivative of $\pi(T)$ at time T_2 (see Appendix A for details),

$$\left.\frac{d^2\pi(T)}{dT^2}\right|_{T=T_2} = \frac{1}{T_2}\Big(p'_{opt}(T_2)\big(\lambda_1'(p_{opt}(T_2))(p_{opt}(T_2) - c - hT_2) + \lambda(T_2)\big) - h\lambda(T_2) + \lambda_2'(T_2)(p_{opt}(T_2) - c - hT_2)\Big).$$

Among the three terms of the right-hand side, the latter two are negative, while the first term is zero, which follows from (16). Therefore, T_2 is a local maximum of the function $\pi(T)$. Since any extreme point is shown to be a local maximum, it must be unique. This proves the first statement of the theorem. □

3.4 Solution algorithm

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The next two sections consider linear, polynomial and exponential effects of price and time on the demand rate. The analysis of each particular case is conducted according to the following algorithm:

Step 1. Assume the price and time effects, $\lambda_1(p)$ and $\lambda_2(t)$.

Step 2. Determine $p_{\max}(t)$ and $\varphi(t)$ from (7) and (16), respectively.

Step 3. Determine T_1 by solving the equation $p_{\max}(t) = c + ht$, and define $p_{opt}(t) = \varphi(t)$ for $t \le T_1$, and $p_{opt}(t) = p_{\max}(t)$ for $t > T_1$.

Step 4. Verify the second-order condition (17).

Step 5. Solve (20) to obtain T_2.

Step 6. Compute T_{opt} from Theorem 1.

Step 7. Compute the optimal order quantity, Q_{opt}, by integrating (12).

Step 8. Obtain $\pi(T_{opt})$ by computing $\Pi(Q_{opt}, p_{opt}(T_{opt}), T_{opt}) / T_{opt}$.

4. Linear effect of price on the demand rate

Let $\lambda_1(p(t))$ be linear, i.e.,

$$\lambda_1(p(t)) = a_1 - b_1 p(t),\ a_1, b_1 > 0. \tag{22}$$

By substituting (22) in (7), we obtain that the maximum price at t is

$$p_{\max}(t) = \frac{1}{b_1}(a_1 + \lambda_2(t)).$$

The value of T_1 is determined from

$$p_{\max}(T_1) = c + hT_1.$$

The second-order conditions are satisfied. From (15) and (16), it follows that for $0 \le t \le T_1$,

$$p_{opt}(t) = \frac{1}{2}\left(p_{\max}(t) + c + ht\right), \tag{23}$$

$$\lambda(t) = \frac{b_1}{2}\left(p_{\max}(t) - c - ht\right), \tag{24}$$

Equation (24) shows that the demand rate steadily decreases, and that it vanishes at $t = T_1$. Equation (23) shows that the inventory holding cost factor is what might cause $p_{opt}(t)$ to grow. When h equals zero, $p_{opt}(t)$ always decreases. The time intervals in which $p_{opt}(t)$ grows are those in which $\lambda_2'(t) > -b_1 h$. Suppose, in particular, that $\lambda_2(t)$ is convex,

$\frac{d^2}{dt^2}\lambda_2(t) > 0$. Then, $p_{opt}(\cdot)$ may have three types of behavior, schematically plotted in Figure 1. It either steadily decreases if $\lambda_2'(t) < -b_1 h$ for $t \in [0, T_1]$, or steadily grows if $\lambda_2'(t) > -b_1 h$ for $t \in [0, T_1]$, or decreases until the time point where $\lambda_2'(t) = -b_1 h$ and grows thereafter.

The optimal replenishment period, T_{opt}, is determined from Theorem 1, and the optimal order quantity, given $p_{opt}(t)$ and T_{opt}, is calculated according to (12) by

$$Q_{opt} = \int_0^{T_{opt}} \left(a_1 - b_1 p_{opt}(t)\right) dt + \int_0^{T_{opt}} \lambda_2(t) dt .$$

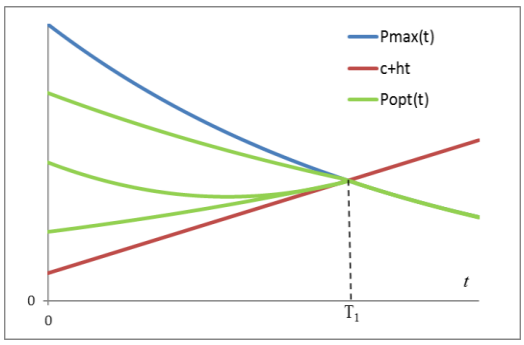

Figure 1. Schematic plot of the functions $p_{\max}(t)$, $c + ht$ and $p_{opt}(t)$.

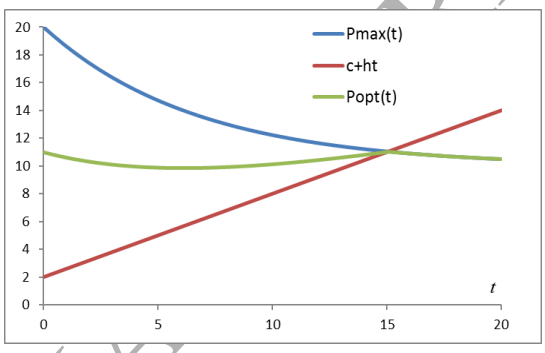

Figure 2. The functions $p_{\max}(t)$, $p_{opt}(t)$, $c + ht$ for the numerical example of negative-exponential time effect on demand.

4.1. Linear time effect

Demand with a linear time effect is modeled as $\lambda(p,t) = a_1 - b_1 p(t) + a_2 - b_2 t$. The reservation and optimal prices are explicitly given, as

$$p_{\max}(t) = \frac{1}{b_1}\left(a_1 + a_2 - b_2 t\right), \quad p_{opt}(t) = \frac{1}{2}\left(h - \frac{b_2}{b_1}\right)t + \frac{1}{2}\left(c + \frac{a_1 + a_2}{b_1}\right), \quad 0 \le t \le T_1,$$

where $T_1 = \frac{a_1 + a_2 - b_1 c}{h b_1 + b_2}$. The $p_{opt}(t)$ function is linear in time, and regardless of whether it increases or decreases, the demand $\lambda(p_{opt}(t), t)$ decays at a rate of $\frac{1}{2}\left(b_1 h + b_2\right)$, and the profit per item decays at a rate of $\frac{1}{2b_1}\left(b_1 h + b_2\right)$. The value of T_2 is obtained from the higher-order polynomial equation (20).

4.2. Negative-exponential time effect

A negative-exponential time effect on demand was considered in You (2005). We generalize that model by not fixing the replenishment time and by allowing the price to continuously change over time (instead of having jumps at discrete time points). The demand is

$$\lambda(p,t) = a_1 - b_1 p(t) + a_2 e^{-b_2 t}. \tag{25}$$

The reservation and optimal prices are

$$p_{\max}(t) = \frac{a_1 + a_2 e^{-b_2 t}}{b_1}, \quad p_{opt}(t) = \frac{1}{2}\left(\frac{a_1 + a_2 e^{-b_2 t}}{b_1} + c + ht\right), \text{ for } 0 \le t \le T_1.$$

Figure 2 shows plots of the functions $p_{\max}(t)$, $p_{opt}(t)$, $c + ht$ for the following parameters:

$$a_1 = 10,\ b_1 = 1,\ a_2 = 10,\ b_2 = 0.15,\ c = 2,\ h = 0.6,\ K = 200,\ T_{\max} = 20.$$

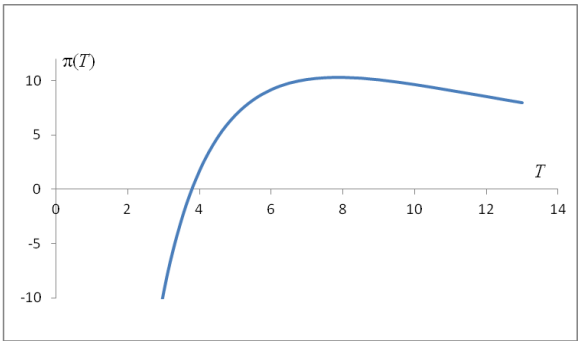

Figure 3. The objective function, $\pi(T)$.

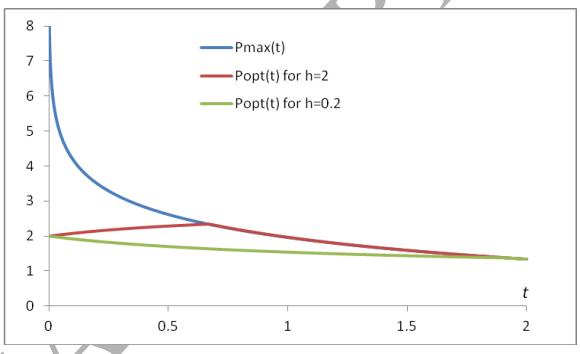

Figure 4. The dynamics of $p_{\max}(t)$ and of $p_{opt}(t)$ for different values of the holding cost *h*.

The figure shows, in particular, that the optimal price decreases for $0 \le t \le 6.1$, then grows up to $t = T_1 = 15$. After that, the demand vanishes, and the optimal price decreases and is equivalent to $p_{\max}(t)$. Figure 3 shows the objective, π, plotted as a function of the replenishment period and with the parameter values used in Figure 2. The optimal replenishment period, determined from (20), is $T_{opt} = T_2 = 8$.

4.3. Polynomial time effect

The polynomial time effect on demand has been studied in Avinadav *et al*. (2013), where the demand model is assumed to be multiplicative rather than additive. Here we consider the following demand rate function:

$$\lambda(p,t) = a_1 - b_1 p(t) + a_2\left(1 - \left(\frac{t}{T_{\max}}\right)^n\right), \quad n > 0.$$

In contrast to the specific linear and negative-exponential time effects discussed above, this polynomial time effect explicitly incorporates the expiration date, $T_{\max}$. The reservation and optimal prices are:

$$p_{\max}(t) = \frac{1}{b_1}\left(a_1 + a_2\left(1 - \left(\frac{t}{T_{\max}}\right)^n\right)\right), \quad p_{opt}(t) = \frac{1}{2b_1}\left(a_1 + a_2\left(1 - \left(\frac{t}{T_{\max}}\right)^n\right)\right) + \frac{c + ht}{2}.$$

5. Nonlinear effect of price on demand rate

According to Petruzzi and Dada (2002), two nonlinear forms of $\lambda_1(p(t))$ commonly appear in the literature: a negative-exponential function, $\lambda_1(p(t)) = ae^{-bp(t)}$, and a power function, $\lambda_1(p(t)) = ap(t)^{-b}$, $a, b > 0$. The negative-exponential demand function was considered by Wang *et al*. (2004) in the context of joint pricing and inventory decisions, and the power function was considered by Hwang and Shinn (1997) in the context of determining the retailer's optimal price and lot size simultaneously when the supplier permits a delay in payments.

For the exponential function, $\lambda_1(p(t)) = a_1 e^{-b_1 p(t)}$, the reservation price is

$$p_{\max}(t) = -\frac{1}{b_1}\ln\left(-\frac{\lambda_2(t)}{a_1}\right),$$

which requires that $-a_1 < \lambda_2(t) < 0$. The optimal price satisfies the nonlinear equation

$$p_{opt}(t) = c + ht + \frac{1}{b_1}\left(1 - e^{-b_1(p_{\max}(t) - p_{opt}(t))}\right). \tag{26}$$

This case extends the case of the linear price effect on demand, considered in Section 4, in which the value of the holding cost is the only factor determining whether $p_{opt}(t)$ increases. Similarly to the linear price effect on demand, the optimal price (26) always decreases for $h = 0$, $\frac{dp_{opt}(t)}{dt} < 0$. This can be shown by differentiating both sides of (26) with $h = 0$,

$$\frac{dp_{opt}(t)}{dt} = e^{-b_1(p_{\max}(t) - p_{opt}(t))}\left(\frac{dp_{\max}(t)}{dt} - \frac{dp_{opt}(t)}{dt}\right), \tag{27}$$

and noting that since $\frac{dp_{\max}(t)}{dt} \le 0$, a positive $\frac{dp_{opt}(t)}{dt}$ cannot satisfy (27). Figure 4 illustrates the dynamics of $p_{opt}(t)$ for the following parameter values: $a_1 = 10$, $b_1 = 1$,

$b_2 = 0.15$, $c = 1$, $\lambda_2(t) = a_1(e^{-b_2 t} - 1)$; and two values for the inventory holding cost, $h = 2$ and $h = 0.2$. As expected, a high (low) value of *h* leads to an increasing (decreasing) optimal price. For the power function, $\lambda_1(p(t)) = a_1 p(t)^{-b_1}$, the reservation price is $p_{\max}(t) = \left(-\frac{\lambda_2}{a_1}\right)^{-\frac{1}{b_1}}$. The optimal price satisfies the nonlinear equation

$$p_{opt}(t) = c + ht + \frac{p_{opt}(t)}{b_1}\left(1 - \left(\frac{p_{opt}(t)}{p_{\max}(t)}\right)^{b_1}\right).$$

6. Discussion

6.1 Implementation issues

The implementation of dynamic pricing in traditional and electronic commerce requires enhanced inventory-handling capabilities and information technologies. Accordingly, recent growth in internet-based marketing has stimulated widespread implementation of dynamic pricing, as a potential means for sellers to increase their total revenues (Levin *et al*. 2010). In offline retail environments, RFID technology provides opportunities to automatically collect real-time data with respect to product quality, and makes it possible to implement dynamic pricing in discrete time as well as in continuous time. Successful applications of RFID technology and RFID-related research in the context of inventory control of perishable products are discussed in Liu *et al*. (2008).

Discrete-time price adjustment policies are probably easier to implement compared with continuous-time adjustment policies. However, a continuous-time policy is more flexible and can lead to a considerable gain in revenue. Additionally, when adopting a continuous-time policy, the retailer does not need to make an arbitrary decision about the frequency of price changes (once per day, hour, etc.). To set a dynamic policy in which the price is changed at given regular intervals, it is possible to discretize the function $p_{opt}(t)$, for example, by averaging its value over the interval of constant price.

6.2. Effect of the expiration duration $T_{\max}$

This section compares the model outlined above with two other models that explicitly account for the expiration date of perishables items, $T_{\max}$, and discusses the implications

regarding dynamic pricing vs. static one as well as additive vs. multiplicative demand function. In particular, we address the case in which the retailer seeks to maximize profit (3) under a demand rate modeled with a linear price effect and a polynomial time effect.

The first model, analyzed in Section 4.3 of this paper, assumes additive demand and a dynamic price:

$$\lambda(p,t) = a_1 - b_1 p(t) + a_2\left(1-\left(\frac{t}{T_{\max}}\right)^n\right) \tag{28}$$

The second model assumes additive demand and a fixed price:

$$\lambda(p,t) = a_1 - b_1 p + a_2\left(1-\left(\frac{t}{T_{\max}}\right)^n\right). \tag{29}$$

This model is a special case of (28). Given a replenishment period, T, the optimal price under the second model is:

$$p_{opt}(T) = \frac{1}{2}\left[c + \frac{1}{2}hT + \frac{a_1}{b_1} + \left(\frac{a_2}{b_1}\left(1-\frac{1}{(n+1)}\left(\frac{T}{T_{\max}}\right)^n\right)\right)\right] \tag{30}$$

The third model, studied in Avinadav et al. (2013), assumes multiplicative demand and a fixed price:

$$\lambda(p,t) = \lambda_0(1-\alpha p)\left(1-\left(\frac{t}{T_{\max}}\right)^n\right). \tag{31}$$

As proven in Avinadav *et al*. (2013), the optimal price in this case is

$$p_{opt}(T) = \frac{\alpha^{-1}+c}{2} + \frac{hT}{2}\frac{n+1}{n+2}\left(1-\frac{n}{2(n+1-(T/T_{\max})^n)}\right). \tag{32}$$

There are two major differences between the additive models (28) and (29) on one hand and the multiplicative model (31) on the other hand. First, the reservation price in the multiplicative model is constant in time, $p_{\max}(t) = \alpha^{-1}$, whereas in the additive models, $p_{\max}(t)$ decreases in time; the latter case is more applicable to the marketing of perishable

products. Second, with respect to the time effect, the multiplicative demand comes to zero at and only at the expiration date, $T_{\max}$, whereas the additive formulations provide a more flexible approach to modeling cases in which demand vanishes as a function of a combination of price and time.

In order to put the comparison of the models in a proper context, it is necessary to select parameter values that reflect actual consumer behavior. In particular, the reservation price, $p = p_{\max}$, and the demand at p = 0 arise from aggregate customers' decisions, rather than depend on the decisions of the retailer. Hence, the parameters of the models have to satisfy:

(1) the reservation price at t = 0, $p_{\max}(0)$, is identical across the three models.

(2) the demand for p = 0 is identical for all t.

For the above conditions to be true, we have

$$a_1 = 0, \quad a_2 = \lambda_0, \quad b_1 = \lambda_0 \alpha . \tag{33}$$

We have derived the optimal policies of the three models in the case characterized by the following parameters: $n = 1$, $K = 2000$, $c = 50$, $h = 1$, $\lambda_0 = 200$, $\alpha^{-1} = 300$, $a_1 = 0$, $a_2 = 200$, $b_1 = 2/3$. The parameters satisfy (33) as described above. The optimal profit per unit of time, π, is depicted in Figure 5 as a function of the expiration time, $T_{\max}$. We observe that the profit increases with $T_{\max}$. The rate of increase can be considered a reflection of the extent to which it is profitable for the manufacturer to invest in extending $T_{\max}$. All three models support the idea that, for smaller values of $T_{\max}$, it is more profitable to invest in extending products' shelf-lives. This is because the rate of increase of π is higher for smaller values of $T_{\max}$ than for larger values of $T_{\max}$.

In the considered case, the additive model with dynamic pricing is associated with a higher profit compared with the additive model with static pricing. Specifically, the former model yields a benefit of about 30% at $T_{\max} = 1$, which drops down to only 2% at $T_{\max} = 5$. The gap between the curve of the multiplicative model and the two curves of the additive models points to the importance of selecting a demand model that accurately represents

customers' behavior. Suppose, for example, that the retailer believes that the demand follows the multiplicative pattern. She then applies the optimal strategy (32) and expects to obtain profit according to the blue line in Figure 5. If, in fact, the demand is additive, the actual profit will be much smaller, bounded from above by the green line in Figure 5. This observation suggests that it is necessary to carry out an empirical study to identify the correct demand pattern associated with perishable products.

6.3. Effect of uncertainty

This section compares the additive and multiplicative models in terms of their sensitivity to the uncertainty of the demand parameters. We consider the following additive and multiplicative demand rate functions:

$$\lambda^{add} = -b_1 p + a e^{-b_2 t}, \qquad \lambda^{m} = \left(-b_1 p + a\right) e^{-b_2 t}$$

These functions satisfy conditions (1) and (2) stated in Section 6.2, indicating that the two models can be compared. The optimal solutions of the additive and multiplicative models are

$$p_{opt}^{add}(t) = \frac{1}{2}\left(\frac{a e^{-b_2 t}}{b_1} + c + ht\right) \text{ and } p_{opt}^{m}(t) = \frac{1}{2}\left(\frac{a}{b_1} + c + ht\right),$$

respectively. For the parameters $a = 30$, $b_1 = 1$, $b_2 = 0.15$, $c = 2$, $h = 0.5$, $K = 200$, $T_{\max} = 5$, the optimal replenishment period, order quantity and profit are summarized in Table 1.

Table 1. Solutions of the additive and multiplicative models.

Model	T_{opt}	Q_{opt}	π_{opt}
Additive	3.34	34.43	50.29
Multiplicative	4.20	41.90	86.48

Suppose now that the parameter a is not perfectly known. Instead, an empirical study has suggested that a is distributed uniformly with a given relative spread (denoted by w), i.e., $a \sim Uniform[30 \cdot (1-w),\ 30 \cdot (1+w)]$. The sensitivity of each of the two models to uncertainty in demand can be measured as the expected profit yielded by the uncertain model relative to

the profit of the deterministic one. We denote this ratio by ρ, and present ρ as a function of w in Figure 6.

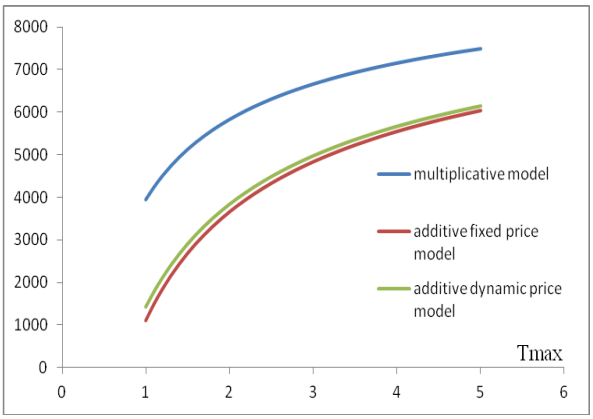

Figure 5. The profit curve for different expiration durations for the three demand models.

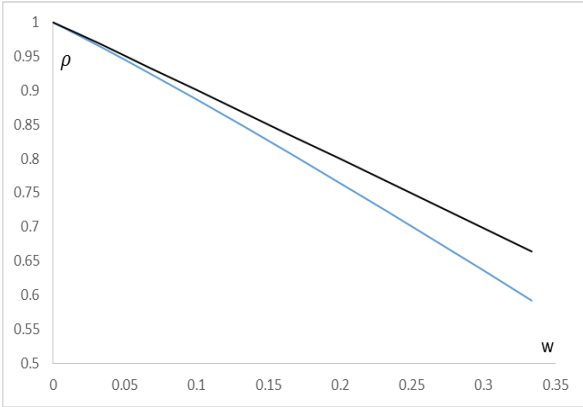

Figure 6. The value of ρ as the spread of the distribution function of a increases: multiplicative model (black line) and additive model (blue line).

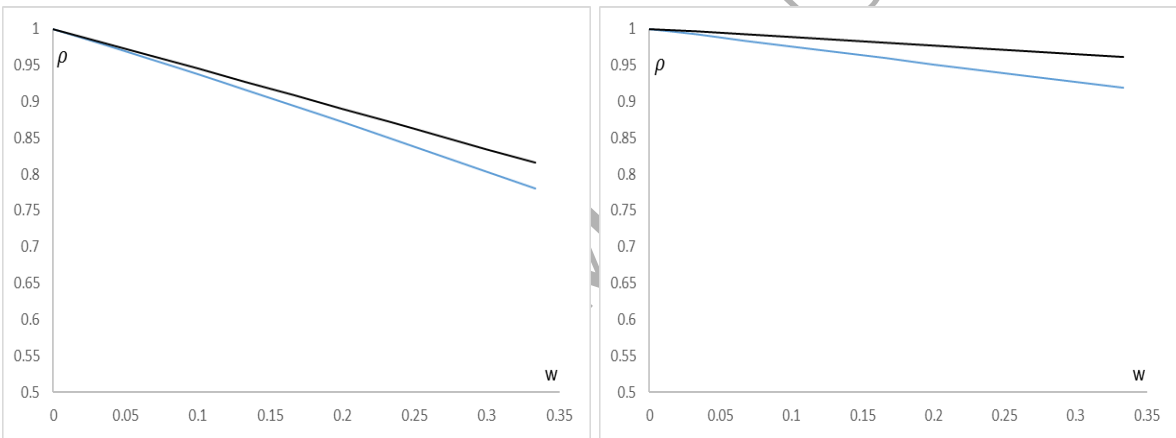

Figure 7. The value of ρ as the spread of the distribution function of b_1 (left side) and b_2 (right side) increases: multiplicative model (black line) and additive model (blue line).

Figure 7 shows the sensitivity with respect to the parameters b_1 and b_2. Figures 6 and 7 indicate that both models are quite sensitive to uncertainty in the demand parameters. The sensitivity to the time factor (parameter a) is higher than that to the price factor (parameter b_1). We observe that for deviations of about 25% in the parameters a and b_1, the profit deteriorates at rates of about 25%-30% and 14%-17%, respectively. The impact of the uncertainty in the decay factor, b_2, is much smaller, at about 5% only. We also observe that, for each of the three parameters, the effect of uncertainty on the seller's outcome is less pronounced under the multiplicative model than under the additive one. The difference

between the models' ρ measure increases as uncertainty increases, and comes to about 5% for a deviation of about 25% in the value of a, b_1 or b_2.

7. Differential pricing policy

In this section we consider a case where the items from the previous replenishment cycle are physically separated from the fresh delivery, and separate pricing policies are set for fresh items and for older products. Fresh and old items are considered as different competitive products, each with its own dynamic pricing policy. The goal is to determine optimal policies for the two products, the length of the replenishment period and the quantities of both products at the beginning of the period. The problem formulation now includes:

$p_1(t)$ and $p_2(t)$ - the selling prices of the fresh and old items, respectively;

$\lambda(p_1(t), p_2(t), t)$ and $\mu(p_1(t), p_2(t), t)$ - the demand rates of the two products;

$I_1(t)$ and $I_2(t)$ - the inventory levels of the two products;

Q and q – the quantity of the two products at the beginning of the replenishment period.

The profit per replenishment period and the inventory dynamic equations are modified as

$$\Pi(Q, q, p_1(t), p_2(t), T) = \int_0^T (p_1(t) - c)\lambda(p_1(t), p_2(t), t)dt + \int_0^T (p_2(t) - c)\mu(p_1(t), p_2(t), t)dt -$$

$$-h\int_0^T (I_1(t) + I_2(t))dt - K, \tag{34}$$

$$\frac{d}{dt}I_1(t) = -\lambda(p_1(t), p_2(t), t), \quad \frac{d}{dt}I_2(t) = -\mu(p_1(t), p_2(t), t), \quad I_1(T) = I_2(0) = q, \quad I_2(T) = 0.$$

Figure 8 schematically presents the functions $I_1(t)$ and $I_2(t)$.

Assumptions 1 and 2 made in Section 2 now apply to both demand rates, λ and μ. Namely, the demand rates separately depend on price and time,

$$\lambda(p_1, p_2, t) = \lambda_1(p_1, p_2) + \lambda_2(t) \text{ and } \mu(p_1, p_2, t) = \mu_1(p_1, p_2) + \mu_2(t) \tag{35}$$

and there exist reservation prices $p_{1,\max}(p_2, t)$ and $p_{2,\max}(p_1, t)$ defined as the lowest prices for the fresh and old items, respectively, at which the product is no longer demanded, given the price of the competitive product and the time, i.e.,

$$\lambda_1(p_{1,\max}, p_2) + \lambda_2(t) = 0, \quad \mu_1(p_1, p_{2,\max}) + \mu_2(t) = 0, \quad t \in [0, T]. \tag{36}$$

The assumptions allow the demand rate to be consistently defined as an additive function (35), as discussed in Section 2. We note that $p_2(t)$ is set at $p_{2,\max}(p_1,t)$ at (and after) the point in time where the old items have run out and only fresh items remain on the shelf.

The properties of the optimal solution proven in Section 3 are partially changed. Specifically, Lemma 2 is no longer valid, since the differential pricing policy, being more flexible, may result in a strictly positive inventory, $q > 0$, at the end of the replenishment period (see Figure 8). The value of q is a new decision variable independent of the other decision variables, $p_1(t)$, $p_2(t)$ and T.

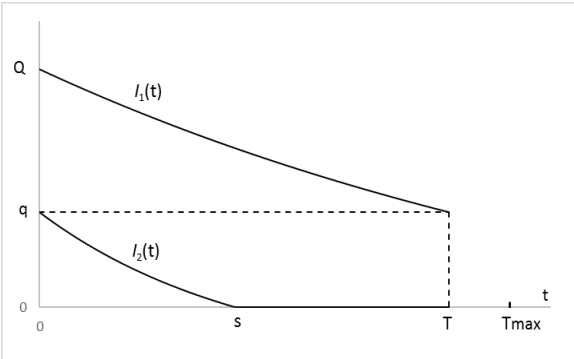

Figure 8. Schematic inventory dynamics of the two products.

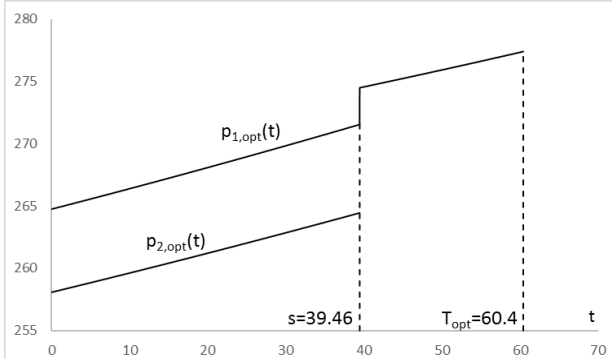

Figure 9. Optimal dynamic pricing for the scenario of Section 7.2.

The application of the maximum principle to the sub-problem of determining the optimal dynamic price, $p_{i,opt}(t)$, $i=1,2$, given the replenishment period now results in the following conditions (similar to (16)),

$$\frac{\partial\lambda_1(p_1,p_2)}{\partial p_1}(p_1(t)-c-ht)+\lambda(p_1,p_2,t)+\frac{\partial\mu_1(p_1,p_2)}{\partial p_1}(p_2(t)-c-h(t+T))=0,$$

$$\frac{\partial\lambda_1(p_1,p_2)}{\partial p_2}(p_1(t)-c-ht)+\mu(p_1,p_2,t)+\frac{\partial\mu_1(p_1,p_2)}{\partial p_2}(p_2(t)-c-h(t+T))=0, \qquad (37)$$

$$\text{if } 0<p_1(t)<p_{1,\max}(p_2(t),t) \text{ and } 0<p_2(t)<p_{2,\max}(p_1(t),t).$$

Contrary to the conclusion of Section 3.2, the shape of $p_{i,opt}(t)$ now does depend on the length of the period, T. This follows since (37) involves T explicitly. A major implication of this new property is that the solution of the sub-problem, which seeks the optimal replenishment period, T_{opt}, becomes significantly more complicated. Since an analog to Theorem 1 is difficult to prove in this case, we determine T_{opt} numerically, as explained in Section 7.1 and illustrated in Section 7.2.

7.1 Numerical solution method

This section presents a method for determining approximate values of the decision variables: dynamic pricing functions $p_{i,opt}(t)$, $i=1,2$, T_{opt} and q_{opt}. The $p_{i,opt}(t)$ functions are obtained from the optimality conditions (37), while the parameters T_{opt} and q_{opt} are determined by exhaustive search. The steps of the method are as follows.

Step 1. Assume the price and time effects, $\lambda_1(p_1,p_2)$, $\mu_1(p_1,p_2)$, $\lambda_2(t)$ and $\mu_2(t)$.

Step 2. Determine $p_{1,\max}(p_2(t),t)$ and $p_{2,\max}(p_1(t),t)$ from (36), and $p_{i,opt}(t)$ from (37).

Step 3. Set q = 0 and $T=T_{\max}$.

Step 4. Integrate $I_2(t)$ from left to right with the initial condition, $I_2(0)=q$, and determine the time, s, at which the inventory of the old items runs out, $I_2(s)=0$ (see Figure 8).

Step 5. Integrate $I_1(t)$ over $t\in[0,T]$ from right to left with the terminal condition, $I_1(T)=q$ and with $p_2(t)=p_{2,\max}(p_1,t)$ for $t>s$ and $p_2(t)=p_{2,opt}(t)$ for $t\le s$.

Step 6. Obtain $Q_{opt}=I_1(0)$.

Step 7. Calculate the objective, $\pi(q,T)$.

Step 8. Repeat Steps 4-7 while decreasing T from $T_{\max}$ to 0 and increasing q from 0 to a value for which one of the inequalities

$$s<T \quad \text{or} \quad s<T_{\max}-T$$

does not hold.

The q and T values that result in the maximum $\pi(q,T)$ yield the optimal solution.

7.2 Example

We have implemented the method presented in the previous section for the case where the price of the products has a linear effect on demand, and time is an exponentially decaying factor. Namely,

$$\lambda(p_1,p_2,t)=ae^{-bt}-b_{11}p_1(t)+b_{12}p_2(t), \quad \mu(p_1,p_2,t)=ae^{-b(t+T)}+b_{21}p_1(t)-b_{22}p_2(t),$$

where $a, b, b_{11}, b_{12}, b_{21}, b_{22}$ are positive parameters. These demand rate functions generalize the case considered in Section 4 and additionally assume that the demand of a product increases with the price of the alternative product. For consistency across cycles, the time factor satisfies $\mu_2(t)=\lambda_2(t+T)$.

Step 2 of the method determines the price functions as follows. For $0\le t<s$,

$$p_{1,opt}(t)=\frac{2b_{22}\lambda_2(t)+(b_{12}+b_{21})\mu_2(t)+(2b_{22}b_{11}-b_{12}(b_{12}+b_{21}))(c+ht)+b_{22}(b_{12}-b_{21})(c+h(t+T))}{4b_{11}b_{22}-(b_{12}+b_{21})^2}$$

,

$$p_{2,opt}(t)=\frac{2b_{11}\mu_2(t)+(b_{12}+b_{21})\lambda_2(t)+(2b_{22}b_{11}-b_{21}(b_{12}+b_{21}))(c+h(t+T))+b_{11}(b_{21}-b_{12})(c+ht))}{4b_{11}b_{22}-(b_{12}+b_{21})^2}$$

.

For $s\le t\le T$, $$p_{1,opt}(t)=\frac{1}{2}\left(c+ht+\frac{b_{22}\lambda_2(t)+b_{12}\mu_2(t)}{b_{11}b_{22}-b_{12}b_{21}}\right).$$

Steps 4-8 computed with the parameters,

$a=0.6, b=0.0015$, $c=20$, $h=1$, $K=2000$,

$b_{11}=0.0025, b_{12}=0.00175, b_{21}=0.00125, b_{22}=0.003$,

identify the optimal replenishment period, $T_{opt}=60.4$, the inventory remaining at the replenishment point, $q_{opt}=3.3$, and the maximum objective, $\pi(q_{opt},T_{opt})=57.04$. Figure 9 presents the optimal dynamic prices. We observe that in this example, the price of the fresh items, $p_{1,opt}(t)$ grows in time; at *t*=*s*, when the items remaining from the previous period run out, $p_{1,opt}(t)$ jumps up; it reaches a maximum at the end of the replenishment period, *t* = *T*; then, the price drops down when a new batch arrives (from now on, the price is denoted by $p_{2,opt}(t)$) and continues to grow until the batch is fully sold at time *t* = *T*+*s*.

8. Conclusions

This paper addresses the problem of determining the optimal dynamic pricing policy and replenishment schedule for a perishable product, with the aim of maximizing the retailer's profit. A demand model that additively combines the effects of price and time contains an inherent contradiction; therefore, we propose a pseudo-additive demand model that resolves this contradiction.

By decomposing the problem, and applying control-theoretic analysis, we obtained necessary optimality conditions for dynamic pricing. We studied the properties of the optimal price $p_{opt}(t)$ and showed that it is independent of the replenishment period, *T*. In contrast, in both additive and multiplicative models that assume a fixed price, the optimal price is dependent on the replenishment period. The optimal dynamic pricing policy $p_{opt}(t)$ can have non-monotonic, complex dynamics, as shown in a number of examples. We further

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developed an algorithm for determining $p_{opt}(t)$ and the optimal replenishment period. These results are discussed in the context of a more general model, where fresh and old items of the same type compete, and different pricing policies must be set to regulate the demand for both.

Specific cases of linear and non-linear effects of price and time on demand were analyzed in order to provide insight into the dynamics of the optimal pricing function under different conditions. In particular, we observed that the value of the holding cost *h* determines whether the optimal price increases or decreases over time. We first assumed a linear price effect and obtained explicit formulas for the optimal pricing policy, under different forms of the time effect on demand. We also analyzed our model under the assumption of a negative-exponential time effect on demand, as in the case considered by You (2005). In contrast to You (2005), who obtained two possible curves for the optimal dynamic price, which were dependent on the value of a pre-determined parameter, we have obtained a single formula for the optimal price over the entire replenishment cycle. The difference between our results and those of You (2005) is explained by the difference between continuous-time and discrete-time representations of price, as well as by the absence of the non-negativity constraint $\lambda(p(t),t) \geq 0$ in You (2005).

A key assumption in the formulation of our problem was the form of demand, i.e., the manner in which price and time affect the demand rate. A comparison of the implications of modeling demand as a multiplicative versus an additive function shows that the retailer's optimal policy significantly differs between the two cases. We believe that a comprehensive statistical study is warranted in order to identify how demand for perishable products should actually be modeled; in particular, it is necessary to identify the cases in which multiplicative versus additive demand formulations reflect customers' behavior. Additional research avenues for generalization of the models proposed herein include consideration of multiple retailers and random demand settings.

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Appendix A

Consider the first derivative of $\pi(T)$ with respect to *T*,

$$\frac{d\pi(T)}{dT}=\frac{d}{dT}\left(\frac{1}{T}\left(\int_0^T (p_{opt}(t)-c)\lambda(p_{opt}(t),t)dt-h\int_0^T I(Q,p_{opt}(t),t)dt-K\right)\right)=$$

$$=-\frac{1}{T^2}\left(\int_0^T (p_{opt}(t)-c)\lambda(p_{opt}(t),t)dt-h\int_0^T I(Q,p_{opt}(t),t)dt-K\right)+$$

$$+\frac{1}{T}\left((p_{opt}(T)-c)\lambda(T)-h\frac{d}{dT}\int_0^T\int_t^T \lambda(\tau)d\tau dt\right)=$$

$$=-\frac{\pi(T)}{T}+\frac{1}{T}\Big((p_{opt}(T)-c)\lambda(T)-hT\lambda(T)\Big)=-\frac{1}{T}\Big(\pi(T)-\lambda(T)(p_{opt}(T)-c-hT)\Big).$$

The second derivative of $\pi(T)$ at T_2 where the first derivative is zero, is

$$\left.\frac{d^2\pi(T)}{dT^2}\right|_{T=T_2}=\left.\frac{d}{dT}\left(-\frac{1}{T}\Big(\pi(T)-\lambda(T)(p_{opt}(T)-c-hT)\Big)\right)\right|_{T=T_2}=$$

$$= -\frac{1}{T_2}\Big(-\lambda'(T_2)(p_{opt}(T_2) - c - hT_2) - \lambda(T_2)(p'_{opt}(T_2) - h)\Big) =$$

$$= \frac{1}{T_2}\Big(\big(\lambda_1'(p_{opt}(T_2))p'_{opt}(T_2) + \lambda_2'(T_2)\big)(p_{opt}(T_2) - c - hT_2) + \lambda(T_2)(p'_{opt}(T_2) - h)\Big) =$$

$$= \frac{1}{T_2}\Big(p'_{opt}(T_2)\big(\lambda_1'(p_{opt}(T_2))(p_{opt}(T_2) - c - hT_2) + \lambda(T_2)\big) - h\lambda(T_2) + \lambda_2'(T_2)(p_{opt}(T_2) - c - hT_2)\Big)$$

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